

Loom Benchmark B1

Pre-registered multi-tier evaluation of the Loom equity research engine. Generated 2026-09-28.

Declared limitations, recorded before the run

These are limitations of the data, not of the analysis, and they are recorded now so they cannot be presented later as findings.

1. Regime coverage is bull-only for document Loom. All three Tier 1 dates sit in a rising 2026 market. Bear-market behaviour of the verdict is untested and untestable with this corpus.
2. The three Tier 1 dates overlap. They are not independent draws. The block bootstrap widens intervals but cannot manufacture independence. Effective sample size is closer to two than three.
3. Findings exist for about 40 companies. Not 1,006. The wide coverage is prices and fundamentals.
4. The reader is Gemini 3.6 Flash, the same model family as Loom's own extraction. A shared-model confound. Partly probed by rerunning a subset on 3.5 Flash, which is a weak proxy for a different reasoning style.
5. Human transferability is not tested. No humans are in this run. A blinded human protocol is produced as an artefact, unrun. Any claim that these effects transfer to human readers is unsupported by this benchmark.
6. Multiplicity. Roughly 30 exploratory comparisons. BH-corrected, and single uncorrected contrasts are not to be read as findings.

How this was implemented

What ran, and what did not

The benchmark has two tiers. The mechanical tier completed in full: 105 point-in-time cross-sections from January 2018 to August 2026, each a rebuild of Loom's factor layer as it would have stood on that month end, with no language model involved anywhere. The reader tier, which measures whether a person using Loom decides better, is quota-bound: the free model tier allows twenty calls per day per model, so it fills a few cells a night and is reported at whatever n it has reached.

Why the tiers are split

Loom's document layer covers about forty companies and begins in October 2025. Its quantitative layer covers hundreds of companies back to 2018. Any claim about regimes, sectors, size strata or multiple independent start dates can therefore only be made about the quantitative layer, and is made only about it here. The reader tier is confined to two 2026 decision dates in a rising market and cannot speak to bear-market behaviour at all.

The company panel, and why it is 400 and not 1,006

Every cross-section is drawn from a fixed panel of the 400 largest companies by SEC rank that have both stored prices and stored fundamentals. The cap is a property of the hardware, not a judgement: this machine has 8GB of RAM, Docker holds 3.8GB of it and the application container already occupies 2.2GB. Loading a thousand companies' bars and facts as ORM objects exhausted free memory and surfaced as 'could not read block 0 ... I/O error', which reads like database corruption and was not. At 400 companies peak resident memory is 1.12GB. This matters for one stated result: the small-cap stratum has only six usable cross-sections, so robustness across company size is weakly tested rather than tested.

How no-lookahead is enforced

Three redundant layers, because a leak is invisible in the output and fatal to the result. First, at the source: findings are filtered by `occurred_at`, factor percentiles are recomputed by the point-in-time scorer rather than read from the stored table, briefs are rebuilt rather than reused, and prices are taken with a strictly-backwards `on_or_before`. Second, by provenance: every record that fed a packet is re-queried and its date asserted against the decision date, independently of the code that selected it. Third, by text: a regex sweep rejects any date later than the decision. Each layer is paired with a deliberate violation in `gates.py` that it must catch, so the audits are known to fire rather than assumed to.

The fundamentals are filing-dated, which was checked and not assumed

The classic backtest leak is treating a fiscal period end as the moment the figure became public. Measured across 728,000 facts, 99.98 percent carry a positive lag between the filing date and the period end, mostly 31 to 60 days for quarterlies and over 120 for annual reports. Only 142 rows are zero-lag or negative. Rows with no period end are dropped by the adapter rather than defaulted, so the future-dated ones never enter scoring.

The portfolios, and what each one isolates

`control_equal` holds the whole cross-section equally weighted and is the no-information baseline. `control_random` holds a random subset of the same size as Loom's book. `loom_long` holds the top quintile by Loom's production composite, equally weighted, so it differs from the control in selection only. `loom_sized` holds the entire universe tilted by percentile, so it differs in sizing only. That pair is the sizing-versus-selection decomposition and it needs no model call. `loom_long_flat` substitutes the plain average of percentiles for the production theme-weighted composite, which is how the theme weighting is attributed. Each individual factor also runs as its own long-short quintile book.

How the effects are decomposed

Every name in the cross-section is classified by what Loom did to it and what then happened: `captured_upside`, `added_loss`, `missed_upside`, `avoided_loss`. Contributions are weighted by the change in exposure rather than by the raw return. An earlier unweighted version counted a name dropped from a 740-name control as heavily as a name held at a twentieth of the book, and the four effects summed to nothing recognisable; weighted, they reconcile to the actual difference between the two books to within a few basis points.

The statistics, written from scratch

This machine has no numpy or scipy, so the statistics are pure Python and validated against known values: the Student-t p-values, computed through a continued-fraction incomplete beta rather than a normal approximation, match scipy to four decimal places. Two conventions are load-bearing. A statistic that cannot be computed returns None and never zero, because a zero t-statistic reads as 'measured, no effect' while an absent one means 'not measured'. And overlapping windows are never fed to a plain t-test: monthly observations of a 63- or 126-session holding period share quarters, so standard errors are Newey-West corrected with lags set to the overlap, and every headline is additionally re-tested on a non-overlapping subsample. Intervals come from a stationary block bootstrap. Families of comparisons carry a Benjamini-Hochberg correction.

Why the OLS column is printed at all

To show how far it misleads. The 126-session alpha has a naive t of 4.97, an autocorrelation-robust t of 2.84, and a t of 1.70 once the overlap is removed. Reporting the first would have claimed a result that is not there, and the three columns are shown together so the reader can see the correction rather than take it on trust.

Pre-registration

The primary endpoint, the secondary hypotheses and the declared limitations were written to PREREGISTRATION.md before any reader book was produced, so the endpoint could not be chosen after seeing results. Everything not named primary or secondary there is exploratory and is reported with the multiplicity correction applied.

Integrity gates

check	result
PASS	text audit rejected a future-dated packet
PASS	provenance audit rejected a future finding
INFO	verdict entropy 0.122 bits over 60 names: {'positive': 1, 'insufficient': 59}
WARN	verdict is near-constant; `full` measures a uniform signal, not Loom discriminating between companies
INFO	packet divergence vs its comparator: degraded=0.150, evidence=0.724, full=0.162, no_contradiction=0.042, no_factor=0.344, no_price=0.190, placebo=0.378, quant_loom=0.581, verdict_only=0.446
INFO	provenance newest signal=2026-06-18 bar=2026-06-25 fact=2026-06-25 decision=2026-06-25

Tier 1 — Loom's quantitative layer, measured mechanically

105 monthly point-in-time cross-sections. No language model is involved in this tier: the factor layer is rebuilt as at each month end and portfolios are formed from its output.

Horizon 21 sessions — 87 cross-sections, median 300 companies ranked

Information coefficient: rank correlation between the signal and forward return in excess of QQQ. Newey-West t, corrected for the overlap that monthly observations of a multi-month holding period create. Starred rows survive Benjamini-Hochberg at 0.05.

signal	n	mean IC	IR	t	p	p (BH)	share > 0
volatility	87	-0.0372	-0.18	-1.60	0.1137	0.4633	0.43
sales_yield	87	0.0354	0.26	2.30	0.0240	0.3601	0.56
cash_flow_yield	87	0.0233	0.17	1.57	0.1212	0.4633	0.51
earnings_yield	87	0.0209	0.17	1.53	0.1300	0.4633	0.51
book_to_price	87	0.0167	0.13	1.25	0.2162	0.4633	0.52
cash_to_assets	87	0.0145	0.14	1.36	0.1763	0.4633	0.54
composite_flat	87	0.0132	0.11	1.03	0.3063	0.5198	0.51
leverage_change	87	-0.0086	-0.13	-1.29	0.2002	0.4633	0.49
cash_conversion	87	-0.0071	-0.10	-1.02	0.3119	0.5198	0.47
net_share_issuance	72	0.0071	0.06	0.49	0.6268	0.9124	0.53
composite_weighted	87	0.0048	0.04	0.38	0.7043	0.9124	0.51
return_on_assets	87	-0.0047	-0.04	-0.35	0.7299	0.9124	0.43
accruals	87	-0.0004	-0.00	-0.04	0.9655	0.9928	0.56
asset_growth	87	-0.0002	-0.00	-0.02	0.9877	0.9928	0.49
momentum	87	-0.0002	-0.00	-0.01	0.9928	0.9928	0.56

Portfolios. Excess is over QQQ for the same window. loom_sized holds the whole universe tilted by percentile and so differs from the control only in sizing; loom_long differs only in selection.

book	n	excess	t	p	vol	maxDD	Sharpe	hit	HHI	names
control_equal	87	-0.0001	-0.03	0.9787	0.178	-0.045	1.90	0.47	0.003	329.7
control_random	87	0.0007	0.18	0.8553	0.182	-0.046	1.89	0.48	0.016	65.5
loom_long	87	-0.0013	-0.29	0.7760	0.157	-0.040	1.94	0.47	0.016	65.5
loom_long_flat	87	-0.0004	-0.09	0.9294	0.163	-0.041	1.96	0.47	0.016	67.1
loom_long_short	87	-0.0208	-2.84	0.0057	0.058	-0.018	-0.57	0.47	0.008	131.1
loom_sized	87	-0.0202	-2.79	0.0064	0.053	-0.017	-0.37	0.47	0.005	329.5

Calibration of the composite. Monotonicity (Spearman of decile against mean return): -0.636. A calibrated score rises monotonically across deciles.

composite decile	n	mean forward excess
0	2908	+0.325%
1	2863	+0.196%
2	2877	+0.053%
3	2854	-0.032%
4	2856	-0.598%
5	2891	+0.210%
6	2862	+0.030%
7	2869	-0.165%
8	2871	-0.222%
9	2830	-0.113%

Universe robustness: the same measurement inside size strata and, below the rule, inside sectors.

stratum	n	mean IC	t	p
mega	87	-0.0076	-0.43	0.6680
mid	87	0.0108	0.89	0.3772
small	6	-0.0096	-0.86	0.4274
Financials	87	-0.0360	-1.43	0.1556
Health Care	87	0.0087	0.41	0.6863
Industrials	87	-0.0108	-0.59	0.5557
Technology	87	0.0226	1.25	0.2138
Consumer Discretionary	6	0.0206	0.33	0.7580
Utilities	6	0.1507	1.35	0.2348
Energy and Materials	6	-0.0226	-0.59	0.5809

effect	n	total (pts)	mean
missed_upside	10877	-261.7	-0.02%
avoided_loss	12103	+262.3	+0.02%
captured_upside	2678	+232.0	+0.09%
added_loss	3023	-242.9	-0.08%

Horizon 63 sessions — 85 cross-sections, median 300 companies ranked

Information coefficient: rank correlation between the signal and forward return in excess of QQQ. Newey-West t, corrected for the overlap that monthly observations of a multi-month holding period create. Starred rows survive Benjamini-Hochberg at 0.05.

signal	n	mean IC	IR	t	p	p (BH)	share > 0
volatility	85	-0.0745	-0.37	-2.39	0.0190	0.1429	0.41
sales_yield	85	0.0584	0.41	2.42	0.0178	0.1429	0.60
cash_flow_yield	85	0.0317	0.25	1.55	0.1256	0.4711	0.55
book_to_price	85	0.0246	0.21	1.27	0.2061	0.4882	0.49
return_on_assets	85	-0.0235	-0.19	-1.27	0.2087	0.4882	0.39
net_share_issuance	70	0.0192	0.15	0.87	0.3857	0.5785	0.49
earnings_yield	85	0.0188	0.17	1.13	0.2604	0.4882	0.51
cash_to_assets	85	0.0186	0.20	1.18	0.2414	0.4882	0.60
cash_conversion	85	-0.0158	-0.25	-1.65	0.1022	0.4711	0.41
leverage_change	85	-0.0102	-0.16	-1.01	0.3161	0.5269	0.45
composite_weighted	85	-0.0057	-0.05	-0.31	0.7606	0.9479	0.41
composite_flat	85	0.0042	0.04	0.24	0.8092	0.9479	0.46
accruals	85	-0.0030	-0.04	-0.23	0.8215	0.9479	0.54
asset_growth	85	-0.0021	-0.02	-0.10	0.9196	0.9578	0.47
momentum	85	0.0014	0.01	0.05	0.9578	0.9578	0.60

Portfolios. Excess is over QQQ for the same window. loom_sized holds the whole universe tilted by percentile and so differs from the control only in sizing; loom_long differs only in selection.

book	n	excess	t	p	vol	maxDD	Sharpe	hit	HHI	names
control_equal	85	0.0030	0.31	0.7594	0.188	-0.082	1.69	0.46	0.003	329.9
control_random	85	0.0022	0.22	0.8285	0.192	-0.083	1.62	0.46	0.016	65.6
loom_long	85	0.0001	0.01	0.9929	0.166	-0.071	1.75	0.45	0.016	65.6
loom_long_flat	85	0.0025	0.20	0.8440	0.172	-0.073	1.83	0.47	0.016	67.2
loom_long_short	85	-0.0629	-3.29	0.0015	0.058	-0.035	-0.73	0.45	0.008	131.2
loom_sized	85	-0.0603	-3.19	0.0020	0.053	-0.032	-0.56	0.46	0.005	329.7

Calibration of the composite. Monotonicity (Spearman of decile against mean return): -0.539. A calibrated score rises monotonically across deciles.

composite decile	n	mean forward excess
0	2843	+1.709%
1	2799	+1.543%
2	2813	-0.092%
3	2790	+0.080%
4	2793	-0.384%
5	2826	-0.302%
6	2798	+0.206%
7	2805	-0.791%
8	2807	-0.418%
9	2767	+0.160%

Universe robustness: the same measurement inside size strata and, below the rule, inside sectors.

stratum	n	mean IC	t	p
mega	85	-0.0255	-0.95	0.3435
mid	85	0.0043	0.25	0.8058
small	6	-0.0128	-0.65	0.5441
Financials	85	-0.0516	-1.24	0.2202
Health Care	85	-0.0034	-0.11	0.9165
Industrials	85	-0.0338	-1.41	0.1636
Technology	85	0.0290	1.01	0.3163
Consumer Discretionary	6	-0.0054	-0.10	0.9263
Utilities	6	0.0224	0.17	0.8722
Energy and Materials	6	-0.0796	-3.57	0.0161

effect	n	total (pts)	mean
missed_upside	10360	-482.2	-0.05%
avoided_loss	12107	+459.1	+0.04%
captured_upside	2520	+439.3	+0.17%
added_loss	3054	-439.2	-0.14%

Horizon 126 sessions — 82 cross-sections, median 300 companies ranked

Information coefficient: rank correlation between the signal and forward return in excess of QQQ. Newey-West t, corrected for the overlap that monthly observations of a multi-month holding period create. Starred rows survive Benjamini-Hochberg at 0.05.

signal	n	mean IC	IR	t	p	p (BH)	share > 0
volatility	82	-0.1036	-0.58	-2.46	0.0161	0.1209	0.23
sales_yield	82	0.0787	0.56	2.52	0.0136	0.1209	0.65
return_on_assets	82	-0.0401	-0.33	-1.39	0.1674	0.4367	0.40
cash_flow_yield	82	0.0392	0.33	1.45	0.1507	0.4367	0.61
net_share_issuance	67	0.0343	0.24	1.09	0.2801	0.5253	0.58
book_to_price	82	0.0335	0.28	1.35	0.1820	0.4367	0.54
cash_to_assets	82	0.0230	0.23	0.91	0.3633	0.6054	0.66
momentum	82	0.0226	0.14	0.66	0.5111	0.7667	0.59
cash_conversion	82	-0.0159	-0.28	-1.47	0.1449	0.4367	0.46
leverage_change	82	-0.0151	-0.25	-1.28	0.2038	0.4367	0.37
earnings_yield	82	0.0087	0.09	0.40	0.6910	0.9422	0.55
composite_flat	82	0.0056	0.05	0.22	0.8258	0.9787	0.49
composite_weighted	82	-0.0015	-0.01	-0.06	0.9550	0.9787	0.49
asset_growth	82	-0.0013	-0.01	-0.04	0.9644	0.9787	0.44
accruals	82	0.0005	0.01	0.03	0.9787	0.9787	0.51

Portfolios. Excess is over QQQ for the same window. loom_sized holds the whole universe tilted by percentile and so differs from the control only in sizing; loom_long differs only in selection.

book	n	excess	t	p	vol	maxDD	Sharpe	hit	HHI	names
control_equal	82	0.0065	0.33	0.7455	0.196	-0.116	1.54	0.45	0.003	330.2
control_random	82	0.0058	0.28	0.7815	0.201	-0.118	1.50	0.45	0.016	65.6
loom_long	82	0.0008	0.03	0.9748	0.174	-0.101	1.62	0.45	0.016	65.6
loom_long_flat	82	0.0048	0.18	0.8543	0.180	-0.105	1.69	0.46	0.016	67.3
loom_long_short	82	-0.1294	-3.21	0.0019	0.057	-0.054	-0.77	0.45	0.008	131.3
loom_sized	82	-0.1245	-3.16	0.0022	0.052	-0.047	-0.61	0.45	0.005	330.0

Calibration of the composite. Monotonicity (Spearman of decile against mean return): -0.624. A calibrated score rises monotonically across deciles.

composite decile	n	mean forward excess
0	2744	+4.102%
1	2703	+4.805%
2	2717	+0.141%
3	2694	+0.551%
4	2697	-0.562%
5	2727	-1.597%
6	2702	+0.335%
7	2709	-1.088%
8	2711	-0.425%
9	2671	+0.093%

Universe robustness: the same measurement inside size strata and, below the rule, inside sectors.

stratum	n	mean IC	t	p
mega	82	-0.0115	-0.33	0.7429
mid	82	0.0013	0.05	0.9572
small	6	-0.0670	-6.73	0.0011
Financials	82	-0.0319	-0.56	0.5776
Health Care	82	0.0178	0.39	0.6982
Industrials	82	-0.0463	-1.26	0.2117
Technology	82	0.0254	0.61	0.5467
Consumer Discretionary	6	0.0601	3.80	0.0126
Utilities	6	-0.0825	-0.96	0.3797
Energy and Materials	6	-0.1232	-3.16	0.0252

effect	n	total (pts)	mean
missed_upside	9694	-721.5	-0.07%
avoided_loss	11999	+659.9	+0.06%
captured_upside	2405	+630.3	+0.26%
added_loss	2977	-620.5	-0.21%

Paired against the control, split by what the market did

loom_long minus control_equal on identical dates, so the market is differenced out. Cross-sections are split into terciles by the benchmark's realised return over the holding period. The regime is assigned after the fact, so this characterises the book's exposure rather than describing a tradeable rule.

horizon	regime	n	benchmark range	loom_long - control	t	p
21	falling	29	-12% to -0%	+0.86 pts	3.57	0.0013
21	flat	29	-0% to +4%	-0.14 pts	-0.52	0.6078
21	rising	29	+4% to +16%	-1.08 pts	-3.76	0.0008
63	falling	28	-22% to +2%	+1.92 pts	3.15	0.0040
63	flat	29	+2% to +9%	-0.62 pts	-1.60	0.1218
63	rising	28	+9% to +30%	-2.14 pts	-5.10	0.0000
126	falling	27	-28% to +8%	+2.24 pts	1.35	0.1874
126	flat	28	+9% to +17%	-1.15 pts	-3.63	0.0012
126	rising	27	+17% to +45%	-2.79 pts	-3.74	0.0009

Beta-adjusted alpha against the equal-weight control

Loom's top-quintile book beat an equal-weight control in every falling market and lost in every rising one, at all three horizons. That is what a lower-beta portfolio does mechanically, so the question is whether anything survives adjusting for the exposure. Monthly observations of a multi-month hold overlap, so the OLS column is reported only to show how far it overstates: 4.97 against 2.84 once corrected and 1.70 on non-overlapping data. Starred rows survive Benjamini-Hochberg over all twelve tests.

horizon	book	n	alpha	beta	t OLS	t NW	p NW	n non-ov	t non-ov	p non-ov	p (BH)
21	loom_long	87	+0.206%	0.829	1.35	1.29	0.2010	87	1.35	0.1800	0.5179
21	loom_long_flat	87	+0.189%	0.883	1.09	1.06	0.2909	87	1.09	0.2796	0.5179
21	loom_long_short	87	+0.129%	-0.166	0.80	0.79	0.4316	87	0.80	0.4270	0.5179
21	loom_sized	87	+0.154%	-0.146	0.97	0.97	0.3336	87	0.97	0.3342	0.5179
63	loom_long	85	+0.884%	0.798	3.42	2.46	0.0160	29	2.39	0.0240	0.0959
63	loom_long_flat	85	+0.527%	0.901	1.88	1.31	0.1925	29	0.94	0.3540	0.5179
63	loom_long_short	85	+0.178%	-0.167	0.67	0.48	0.6335	29	0.39	0.7001	0.6335
63	loom_sized	85	+0.374%	-0.156	1.43	1.02	0.3104	29	0.68	0.5012	0.5179
126	loom_long	82	+1.837%	0.796	4.97	2.84	0.0057	14	1.70	0.1127	0.0679
126	loom_long_flat	82	+0.613%	0.933	1.55	0.84	0.4013	14	-0.03	0.9752	0.5179
126	loom_long_short	82	+0.519%	-0.193	1.27	0.64	0.5215	14	-0.05	0.9631	0.5689
126	loom_sized	82	+0.686%	-0.165	1.85	0.90	0.3696	14	0.30	0.7687	0.5179

Tier 2 — reader books

1 books scored. Every contrast is paired inside a (persona, seed, decision date) cell; nothing is averaged across unpaired cells.

arm vs control	metric	n	mean	t	p	CI low	CI high	n positive
----------------	--------	---	------	---	---	--------	---------	------------

Evidence utilisation. A citation counts as grounded only if it appears verbatim in that book's own packet; an ungrounded citation is a confabulation and cannot be read as use of the research.

arm	books	citations	verbatim rate	positions citing
control	1	62	1.000	1.000

Calibration of stated conviction against realised outcome.

arm	n	Brier	ECE	discrimination
control	18	0.2524	0.0900	0.246

Research efficiency.

arm	latency s	input tokens	output tokens
control	56.0	4093	5424